

Stony Brook University
Instructor: Alejandro Melo Ponce

Spring 2018

FINAL EXAM

NAME: _____

ID: _____

Instructions: The exam has 40 questions. Each question has a value of 2.5 points for a total of 100 points. Answer all questions in the scantron provided. Make sure that you fill the bubbles properly. Calculators are allowed but other electronic devices or outside materials are NOT allowed.

You are **NOT** allowed to leave the room until you submit your exam.

GOOD LUCK!

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Multiple Choice Questions

1. Analyzing the behavior of unemployment rates across U.S. states in March of 2006 is an example of using
 - a) time series data.
 - b) panel data.
 - c) cross-sectional data.
 - d) experimental data.
2. Ideal randomized controlled experiments in economics are
 - a) often performed in practice.
 - b) often used by the Federal Reserve to study the effects of monetary policy.
 - c) useful because they give a definition of a causal effect.
 - d) sometimes used by universities to determine who graduates in four years rather than five.
3. In econometrics, we typically do not rely on exact or finite sample distributions because
 - a) we have approximately an infinite number of observations (think of re-sampling).
 - b) variables typically are normally distributed.
 - c) the covariances of Y_i, Y_j are typically not zero.
 - d) asymptotic distributions can be counted on to provide good approximations to the exact sampling distribution (given the number of observations available in most cases).
4. The probability of an outcome
 - a) is the number of times that the outcome occurs in the long run.
 - b) equals $M \times N$, where M is the occurrences and N is the sample size.
 - c) is the proportion of times that the outcome occurs in the long run.
 - d) equals the sample mean divided by the sample standard deviation.
5. Two random variables X and Y are independently distributed if all of the following conditions hold, with the exception of
 - a) $\Pr(Y = y|X = x) = \Pr(Y = y)$.
 - b) knowing the value of one of the variables provides no information about the other.
 - c) if the conditional distribution of Y given X equals the marginal distribution of Y .
 - d) $E(Y) = E[E(Y)]$.
6. An estimator $\hat{\mu}$ of the population value μ is more efficient when compared to another estimator $\bar{\mu}$, if
 - a) $E[\hat{\mu}] > E[\bar{\mu}]$.
 - b) it has a smaller variance.
 - c) its c.d.f. is flatter than that of the other estimator.
 - d) both estimators are unbiased, and $V[\hat{\mu}] < V[\bar{\mu}]$.
7. A type I error is
 - a) always less than one.
 - b) the error you make when rejecting the null hypothesis when it is true.
 - c) the error you make when rejecting the alternative hypothesis when it is true.
 - d) always 5%.

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8. The t -statistic has the following distribution:
 - a) standard normal distribution for $n < 15$
 - b) Student t distribution when the sample is small.
 - c) Student t distribution if the Y is normally distributed and n is small.
 - d) a standard normal distribution if the sample standard deviation goes to zero.
9. The reason why estimators have a sampling distribution is that
 - a) economics is not a precise science
 - b) individuals respond differently to incentives
 - c) in real life you typically get to sample many times
 - d) the values of the explanatory variable and the error term differ across samples.
10. Finding a small value of the p -value (e.g. less than 5%)
 - a) indicates evidence in favor of the null hypothesis
 - b) implies that the t -statistic is less than 1.96.
 - c) indicates evidence against the null hypothesis
 - d) will only happen roughly one in twenty samples.
11. When the estimated slope coefficient in the simple regression model, $\hat{\beta}_1$, is zero, then
 - (a) $R^2 = 0$
 - (b) $0 < R^2 < 1$.
 - (c) $R^2 = \bar{Y}$.
 - (d) $R^2 > \frac{SSR}{TSS}$
12. The following are all least squares assumptions with the exception of:
 - (a) The explanatory variable in the regression model is normally distributed.
 - (b) The conditional distribution of u_i given X_i has a mean of zero.
 - (c) Large outliers are unlikely.
 - (d) $(X_i, Y_i), i = 1, \dots, n$ are independently and identically distributed.
13. The OLS estimator is derived by
 - (a) minimizing the sum of absolute residuals.
 - (b) minimizing the sum of squared residuals.
 - (c) making sure that the standard error of the regression equals the standard error of the slope estimator.
 - (d) connecting the Y_i corresponding to the lowest X_i observation with the Y_i corresponding to the highest X_i observation.
14. In the simple linear regression model, the regression slope
 - (a) when multiplied with the explanatory variable will give you the predicted Y .
 - (b) indicates by how many units Y increases, given a one unit increase in X .
 - (c) represents the elasticity of Y on X .
 - (d) indicates by how many percent Y increases, given a one percent increase in X .

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15. The slope estimator, $\hat{\beta}_1$, has a smaller standard error, other things equal, if
- (a) the sample size is smaller.
 - (b) there is more variation in the explanatory variable, X .
 - (c) the intercept, β_0 , is small.
 - (d) there is a large variance in the error term, u .
16. The t -statistic is calculated by dividing
- (a) the slope by the standard deviation of the explanatory variable.
 - (b) the slope by 1.96.
 - (c) the estimator minus its hypothesized value by the standard error of the estimator.
 - (d) the OLS estimator by its standard error.
17. If the absolute value of your calculated t -statistic exceeds the critical value from the standard normal distribution, you can
- (a) safely assume that your regression results are significant.
 - (b) conclude that most of the actual values are very close to the regression line.
 - (c) reject the null hypothesis.
 - (d) reject the assumption that the error terms are homoskedastic.
18. Finding a small value of the p -value (e.g. less than 5%)
- (a) will only happen roughly one in twenty samples.
 - (b) indicates evidence in against the null hypothesis.
 - (c) implies that the t -statistic is less than 1.96.
 - (d) indicates evidence in favor of the null hypothesis.
19. One of the following steps is not required as a step to test for the null hypothesis:
- (a) compute the p -value.
 - (b) test for the errors to be normally distributed.
 - (c) compute the standard error of $\hat{\beta}_1$
 - (d) compute the t -statistic.
20. You extract approximately 5,000 observations from the Current Population Survey (CPS) and estimate the following regression function:

$$\widehat{ahe} = \underset{(1.00)}{3.32} - \underset{(0.04)}{0.45} \times Age, \quad R^2 = 0.02, \quad SER = 8.66$$

where ahe is average hourly earnings, and Age is the individual's age. Given the specification, your 95% confidence interval for the effect of changing age by 5 years is approximately

- (a) cannot be determined given the information provided.
- (b) [\$2.32, \$4.32]
- (c) [\$1.35, \$5.30]
- (d) [\$1.96, \$2.54]

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21. Under imperfect multicollinearity
- (a) the OLS estimator cannot be computed.
 - (b) two or more of the regressors are highly correlated.
 - (c) the OLS estimator is biased even in samples of $n > 100$.
 - (d) the error terms are highly, but not perfectly, correlated.
22. In multiple regression, the R^2 increases whenever a regressor is
- (a) added unless the coefficient on the added regressor is exactly zero.
 - (b) added.
 - (c) added unless there is heteroskedasticity.
 - (d) greater than 1.96 in absolute value.
23. If you had a two regressor regression model, then omitting one variable which is relevant
- (a) will have no effect on the coefficient of the included variable if the correlation between the excluded and the included variable is negative.
 - (b) will always bias the coefficient of the included variable upwards.
 - (c) can result in a negative value for the coefficient of the included variable, even though the coefficient will have a significant positive effect on Y if the omitted variable were included.
 - (d) makes the sum of the product between the included variable and the residuals different from 0.
24. Perfect multicollinearity
- (a) implies that it will be difficult to estimate precisely one or more of the partial effects using the data at hand
 - (b) violates one of the four Least Squares assumptions in the multiple regression model
 - (c) suggests that a standard spreadsheet program does not have enough power to estimate the multiple regression model
 - (d) happens when you omit one dummy variable.
25. If you reject a joint null hypothesis using the F -test in a multiple hypothesis setting, then
- (a) a series of t -tests may or may not give you the same conclusion.
 - (b) the regression is correctly specified.
 - (c) all of the individual hypotheses are always simultaneously rejected.
 - (d) the F -statistic must be negative.
26. When there are omitted variables in the regression, which are determinants of the dependent variable, then
- (a) you cannot measure the effect of the omitted variable, but the estimator of your included variable(s) is (are) unaffected.
 - (b) the OLS estimator is biased if the omitted variable is correlated with the included variable.
 - (c) this has no effect on the estimator of your included variable because the other variable is not included.
 - (d) this will always bias the OLS estimator of the included variable.

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27. In the multiple regression model, the least squares estimator is derived by
- (a) minimizing the sum of squared prediction mistakes.
 - (b) setting the sum of squared errors equal to zero.
 - (c) minimizing the absolute difference of the residuals.
 - (d) forcing the smallest distance between the actual and fitted values.
28. To test joint linear hypotheses in the multiple regression model, you need to
- (a) compare the sums of squared residuals from the restricted and unrestricted model.
 - (b) use several t-statistics and perform tests using the standard normal distribution.
 - (c) use the heteroskedasticity-robust F-statistic.
 - (d) compare the adjusted R^2 for the model which imposes the restrictions, and the unrestricted model.
29. The dummy variable trap is an example of
- (a) imperfect multicollinearity.
 - (b) something that is of theoretical interest only.
 - (c) perfect multicollinearity.
 - (d) something that happens every time you use dummy variables.
30. The overall regression F -statistic tests the null hypothesis that
- (a) the slope coefficients are zero.
 - (b) all slope coefficients and the intercept are zero.
 - (c) the intercept in the regression and at least one, but not all, of the slope coefficients is zero.
 - (d) the slope coefficient of the variable of interest is zero, but that the other slope coefficients are not.
31. A 95% confidence set for two or more coefficients is a set that contains
- (a) the same values as the 95% confidence intervals constructed for the coefficients.
 - (b) integer values only.
 - (c) the population values of these coefficients in 95% of randomly drawn samples.
 - (d) the sample values of these coefficients in 95% of randomly drawn samples.
32. When there are two coefficients, the resulting confidence sets are
- (a) squares
 - (b) rectangles
 - (c) ellipses
 - (d) trapezoid
33. Consider a regression with two variables, in which X_{1i} is the variable of interest and X_{2i} is the control variable. Conditional mean independence requires
- (a) $E[u_i|X_{1i}, X_{2i}] = E[u_i|X_{2i}]$
 - (b) $E[u_i|X_{1i}, X_{2i}] = E[u_i|X_{1i}]$
 - (c) $E[u_i] = E[u_i|X_{2i}]$
 - (d) $E[u_i|X_{1i}] = E[u_i|X_{2i}]$

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34. The interpretation of the slope coefficient in the model $Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i$ is as follows:
- (a) a change in X by one unit is associated with a β_1 change in Y .
 - (b) a 1% change in X is associated with a $\beta_1\%$ change in Y .
 - (c) a change in X by one unit is associated with a $\beta_1 100\%$ change in Y .
 - (d) a change in 1% is associated with a change in Y of $0.01\beta_1$.
35. For the polynomial regression model,
- (a) you can still use OLS estimation techniques, but the t -statistics do not have an asymptotic normal distribution.
 - (b) the critical values from the normal distribution have to be changed to 1.96^2 , 1.96^3 , etc.
 - (c) you need new estimation techniques since the OLS assumptions do not apply any longer.
 - (d) the techniques for estimation and inference developed for multiple regression can be applied.
36. To test whether or not the population regression function is linear rather than a polynomial of order r ,
- (a) use the test of $(r - 1)$ restrictions using the F -statistic.
 - (b) check whether the regression R^2 for the polynomial regression is higher than that of the linear regression.
 - (c) look at the pattern of the coefficients: if they change from positive to negative to positive, etc., then the polynomial regression should be used.
 - (d) compare the TSS from both regressions.
37. In the regression model $Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 X_i \times D_i + u_i$ where X is a continuous variable and D is a binary variable, the coefficient β_3
- (a) indicates the slope of the regression when $D_i = 1$.
 - (b) indicates the difference in the slopes of the two regressions.
 - (c) has a standard error that is not normally distributed even in large samples since D is not a normally distributed variable.
 - (d) has no meaning since $X_i \times D_i = 0$ when $D_i = 0$.
38. In the model $Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 \times X_2 + u_i$, the expected effect $\frac{\Delta Y}{\Delta X_1}$ is
- (a) $\beta_1 + \beta_3 X_1$.
 - (b) $\beta_1 + \beta_3 X_2$.
 - (c) $\beta_1 + \beta_3$.
 - (d) β_1 .

39. Assume that you had estimated the following regression model

$$\widehat{TestScore} = 607.3 + 3.85Income - 0.0423Income^2.$$

If income increased from 10 to 11 (\$10,000 to \$11,000), then the predicted effect on test scores would be

- (a) 2.96
- (b) 3.85
- (c) $3.85 - 0.0423$
- (d) Cannot be calculated because the function is non-linear.

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40. Consider the population regression of log earnings against two binary variables: whether a worker is married $married_i$, where $married_i = 1$ if the i -th person is married; and the worker's gender $gender_i$, where $gender_i$ if the i -th person is female and an interaction term of the two binary variables. Thus, the model looks like

$$\ln(earnings)_i = \beta_0 + \beta_1 married_i + \beta_2 gender_i + \beta_3 married_i \times gender_i + u_i.$$

The interaction term:

- (a) allows the population effect on log earnings of being married to depend on gender.
- (b) cannot be estimated without the presence of a continuous variable.
- (c) does not make sense since it could be zero for married males.
- (d) indicates the effect of being married on log earnings.